

Coordinate Geometry

ANSWER KEY



Distance and midpoint formulas

- Find the distance between $(2, -1)$ and $(10, 5)$.
Distance $= \sqrt{(10 - 2)^2 + (5 - (-1))^2} = \sqrt{64 + 36} = \sqrt{100} = 10$.
 - Find the midpoint of the segment connecting $(-4, 7)$ and $(6, -3)$.
Midpoint $= \left(\frac{-4 + 6}{2}, \frac{7 + (-3)}{2} \right) = (1, 2)$.
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Slope and equations of lines

- Find the slope of the line through $(1, 4)$ and $(5, -8)$.
Slope $= \frac{-8 - 4}{5 - 1} = \frac{-12}{4} = -3$.
 - Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$, in slope-intercept form.
 $y - (-3) = \frac{1}{2}(x - 2)$, so $y + 3 = \frac{1}{2}x - 1$, giving $y = \frac{1}{2}x - 4$.
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Parallel and perpendicular lines

- Find the equation of the line parallel to $y = 4x - 1$ passing through $(0, 6)$.
Parallel lines share the same slope: $m = 4$. Since $(0, 6)$ is the y -intercept, the equation is $y = 4x + 6$.
 - Find the equation of the line perpendicular to $y = -\frac{2}{3}x + 5$ passing through $(4, 1)$.
The perpendicular slope is the negative reciprocal: $m = \frac{3}{2}$. $y - 1 = \frac{3}{2}(x - 4)$, so
 $y = \frac{3}{2}x - 6 + 1 = \frac{3}{2}x - 5$.
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The equation of a circle

- Write the equation of a circle with center $(3, -2)$ and radius 7.
 $(x - 3)^2 + (y + 2)^2 = 49$.
 - Find the center and radius of the circle $x^2 + y^2 - 6x + 4y - 3 = 0$, by completing the square.
Group and complete the square: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 3 + 9 + 4$, giving
 $(x - 3)^2 + (y + 2)^2 = 16$. Center: $(3, -2)$. Radius: $\sqrt{16} = 4$.
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Reflections, translations, and symmetry

- 9 Find the coordinates of the point $(5, -3)$ after a reflection over the x -axis, then after a reflection over the y -axis.
Reflection over the x -axis: $(5, 3)$ (the y -coordinate's sign flips). Reflection over the y -axis (applied to the original point): $(-5, -3)$ (the x -coordinate's sign flips).
- 10 A point (a, b) is translated 3 units right and 5 units down. Write the coordinates of the image point.
 $(a + 3, b - 5)$.
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Finding area using the coordinate plane

- 11 Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, and $(0, 4)$.
Since the triangle has a right angle at the origin, with legs along the axes: $\text{Area} = \frac{1}{2}(6)(4) = 12$.
- 12 A rectangle has vertices $(1, 1)$, $(1, 5)$, $(7, 5)$, and $(7, 1)$. Find its area and perimeter.
Width: $7 - 1 = 6$. Height: $5 - 1 = 4$. $\text{Area} = 6 \times 4 = 24$. $\text{Perimeter} = 2(6 + 4) = 20$.
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Determining if points are collinear

- 13 Determine whether the points $(1, 2)$, $(3, 6)$, and $(5, 10)$ are collinear, by comparing slopes.
Slope from $(1, 2)$ to $(3, 6)$: $\frac{6-2}{3-1} = \frac{4}{2} = 2$. Slope from $(3, 6)$ to $(5, 10)$: $\frac{10-6}{5-3} = \frac{4}{2} = 2$. Since both slopes are equal, the three points are collinear.
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Finding the intersection of two lines

- 14 Find the point of intersection of $y = 2x - 3$ and $y = -x + 9$.
Set equal: $2x - 3 = -x + 9$, so $3x = 12$, giving $x = 4$. Then $y = 2(4) - 3 = 5$. Intersection: $(4, 5)$.
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Combining coordinate geometry concepts

- 15 This question has two parts. Triangle ABC has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 6)$. (a) Find the length of each side of the triangle. (b) Determine whether the triangle is isosceles, and justify your answer.
(a) $AB = \sqrt{(4-0)^2 + (0-0)^2} = 4$. $AC = \sqrt{(2-0)^2 + (6-0)^2} = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10}$.
 $BC = \sqrt{(2-4)^2 + (6-0)^2} = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10}$. (b) Since $AC = BC = 2\sqrt{10}$ (two sides are equal in length), the triangle is isosceles.
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Graphing inequalities in the coordinate plane

- 16** Determine whether the point $(3, 1)$ satisfies the inequality $y < 2x - 3$.
Substitute: $1 < 2(3) - 3 = 3$? True. The point satisfies the inequality.
- 17** A region is defined by $y \geq x + 1$. Determine whether $(0, 0)$ lies in this region.
Substitute: $0 \geq 0 + 1 = 1$? False. The point $(0, 0)$ does not lie in the region.
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The equation of a parabola in the coordinate plane

- 18** Write the equation of a parabola with vertex $(2, -3)$ that passes through $(4, 5)$.
Vertex form: $y = a(x - 2)^2 - 3$. Substitute $(4, 5)$: $5 = a(4 - 2)^2 - 3$, so $5 = 4a - 3$, giving $a = 2$.
Equation: $y = 2(x - 2)^2 - 3$.
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Finding points that divide a segment in a given ratio

- 19** Find the point that divides the segment from $(1, 2)$ to $(9, 10)$ in a ratio of $1 : 3$ (closer to the first point).
Using the section formula for a $1 : 3$ ratio: $x = 1 + \frac{1}{4}(9 - 1) = 1 + 2 = 3$. $y = 2 + \frac{1}{4}(10 - 2) = 2 + 2 = 4$.
Point: $(3, 4)$.
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A multi-part problem combining lines and circles

- 20** This question has two parts. A circle has equation $(x - 2)^2 + (y + 1)^2 = 25$. (a) State the center and radius. (b) Determine whether the point $(5, 3)$ lies inside, on, or outside the circle, by computing its distance from the center.
(a) Center: $(2, -1)$. Radius: $\sqrt{25} = 5$. (b) Distance from center to $(5, 3)$:
 $\sqrt{(5 - 2)^2 + (3 - (-1))^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Since this distance equals the radius, the point $(5, 3)$ lies exactly ON the circle.
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Systems of a line and a circle

- 21** Find the points of intersection of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$.
Substitute: $x^2 + (x + 1)^2 = 25$, so $x^2 + x^2 + 2x + 1 = 25$, giving $2x^2 + 2x - 24 = 0$, or $x^2 + x - 12 = 0$.
Factor: $(x + 4)(x - 3) = 0$, so $x = -4$ or $x = 3$. Corresponding y -values: for $x = -4$, $y = -3$; for $x = 3$, $y = 4$. Intersection points: $(-4, -3)$ and $(3, 4)$.